

Practice 1. Reading & Converting Formats.

Maximum score: 1 point

Defense: mandatory

Format: individual work

Objective: to select a dataset, analyze its structure and quality, prepare the data, compare storage formats.

Task 1. Dataset selection.

Select an open dataset from Kaggle, Data.gov, World Bank Open Data, UN Data or another open source. The dataset must contain at least 100 rows, five columns and at least one data-quality problem.

- dataset name and source link;
- author or organization;
- number of rows and columns;
- meaning of one row and each column;
- dataset type: structured, semi-structured or unstructured.

Task 2. Data-quality analysis

Identify the data types, missing values, duplicates, invalid values, inconsistent categories, incorrect date or number formats, and possible outliers. Present the results in a table containing the column name, identified problem, example, and its possible effect on further data analysis.

Task 3. Data Formats

Save the prepared dataset in one suitable format and one alternative format. Compare CSV, JSON, Parquet, ORC, and Avro according to their representation, storage structure, schema support, human readability, typical applications, advantages, and limitations. Explain which format is the most appropriate for the selected dataset and justify your choice.

4. Task 4. Data Format Comparison

Compare CSV, JSON, Parquet, ORC, and Avro in terms of data representation, storage structure, schema support, human readability, compression, typical applications, advantages, and limitations. Present the comparison on a table. For the two formats used in your practical work, measure and compare file size, writing time, reading time, preservation of rows and columns, data types, and missing values.

Prepare writing report describing all steps and results and submitting the ipynb file together with the report.